

Alessandro Giacchetto

ETH Fellow, Hermann-Weyl-Instructor · ETHZ, HG J 16.2

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Employment

- Oct 2023–present ETH Fellow & Hermann-Weyl-Instructor, ETH Zürich
Mentor: Rahul Pandharipande
- Oct 2021–Oct 2023 Postdoctoral Fellow, IPhT
Mentor: Bertrand Eynard

Education

- 2021 PhD in Mathematics, MPIM
[*Geometric and topological recursion and invariants of the moduli space of curves*](#)
Supervisor: Gaëtan Borot
- 2017 MSc in Mathematics, University of Trieste & SISSA
[*The J-equation on Kähler manifolds and blowups*](#)
Supervisor: Jacopo Stoppa
- 2015 BSc in Mathematics, University of Trieste
[*Skein relations and polynomial invariants of knots and links*](#)
Supervisor: Bruno Zimmermann

Publications and preprints

- [Theta classes: generalized topological recursion, integrability and \$\mathcal{W}\$ -constraints](#) V. Bouchard, N. K. Chidambaram, S. Shadrin Preprint 2025  [arXiv](#)
- [Les Houches lecture notes on moduli spaces of Riemann surfaces](#) D. Lewański Les Houches Summer School Lecture Notes - 2024-08: Quantum Geometry *SciPost Phys. Lect. Notes* 2025  [arXiv](#)  [Open access](#)
- [The factorial growth of topological recursions](#) G. Borot, B. Eynard *Lett. Math. Phys.* 115.62 2025  [arXiv](#)  [DOI](#)

- [Can transformers do enumerative geometry?](#) B. Hashemi, R. G. Corominas *13th Int. Conf. Learn. Represent.* 2025 [arXiv](#) [Open access](#) [Code](#)
- [Symmetries of F-cohomological field theories and F-topological recursion](#) G. Borot, G. Umer *Commun. Math. Phys.* 406.248 2025 [arXiv](#) [DOI](#)
- [Length spectrum of large genus random metric maps](#) S. Barazer, M. Liu *Forum Math. Sigma* 13 2025 [arXiv](#) [DOI](#)
- [Resurgent large genus asymptotics of intersection numbers](#) B. Eynard, E. Garcia-Failde, P. Gregori, D. Lewański Preprint 2023 [arXiv](#)
- [The spin Gromov–Witten/Hurwitz correspondence for \$\mathbb{P}^1\$](#) R. Kramer, D. Lewański, A. Sauvaget *J. Eur. Math. Soc.* 2025 [arXiv](#) [DOI](#)
- [Relations on \$\overline{\mathcal{M}}_{g,n}\$ and the negative \$r\$ -spin Witten conjecture](#) N. K. Chidambaram, E. Garcia-Failde *Invent. Math.* 241 2025 [arXiv](#) [DOI](#)
- [Shifted Witten classes and topological recursion](#) S. Charbonnier, N. K. Chidambaram, E. Garcia-Failde *Trans. Amer. Math. Soc.* 377.2 2024 [arXiv](#) [DOI](#)
- [An intersection-theoretic proof of the Harer–Zagier formula](#) D. Lewański, P. Norbury *Algebraic Geom.* 10.2 2023 [arXiv](#) [DOI](#)
- [Around the combinatorial unit ball of measured foliations on bordered surfaces](#) G. Borot, S. Charbonnier, V. Delecroix, C. Wheeler *Int. Math. Res. Not.* 2023.17 2023 [arXiv](#) [DOI](#)
- [A new spin on Hurwitz theory and ELSV via theta characteristics](#) R. Kramer, D. Lewański Accepted in *Selecta Math.* 2025 [arXiv](#)
- [On the Kontsevich geometry of the combinatorial Teichmüller space](#) J. E. Andersen, G. Borot, S. Charbonnier, D. Lewański, C. Wheeler Preprint 2020 [arXiv](#)
- [Masur–Veech volumes and intersection theory, the principal strata of quadratic differentials](#) G. Borot, D. Lewański Appendix in *Duke Math. J.* 172.9 2023 [arXiv](#) [DOI](#)
- [Topological recursion for Masur–Veech volumes](#) J. E. Andersen, G. Borot, S. Charbonnier, V. Delecroix, D. Lewański, C. Wheeler *J. London Math. Soc.* 107.1 2023 [arXiv](#) [DOI](#)

Teaching

- [Toric Geometry](#), lecturer, ETHZ, Spring 2025
- *Moduli spaces of Riemann surfaces*, PhD mini-course, Les Houches, Summer 2024
- [Riemann Surfaces](#), lecturer, ETHZ, Spring 2024
- *Hurwitz numbers and Gromov–Witten theory*, PhD mini-course, Hiddensee, Summer 2022

- *Topology*, teaching assistant, UniTS, Spring 2017
- *Linear algebra*, teaching assistant, UniTS, Fall 2016

Organization of events

- Workshop [Refinement in Enumerative Geometry and Physics](#), Bangalore (22 Jun–3 Jul, 2026)
- School [Quantum Geometry](#), Les Houches (29 Jul–23 Aug, 2024)
- School [Topological recursion and integrability](#), Trieste (11–16 Sep, 2023)
- Conference [Moduli spaces: theory and coding](#), Les Diablerets (27 Feb–3 Mar, 2023)
- School + workshop [TR Salento 2022](#), Otranto (5–16 Sep, 2022)
- Conference [Recent advances on moduli spaces of curves](#), Leysin (18–24 Mar, 2022)
- Reading group [Donaldson–Thomas invariants](#), MPIM (Spring 2020)
- Reading group [Integer points in polyhedra](#), MPIM (Fall 2019)
- Reading group [Geometric recursion](#), MPIM (Fall 2018)

Honours

- Oct 2023–Sep 2025 ETH Fellowship
- 2018 Marco Reni prize
- Sep 2015–Sep 2017 SISSA MSc Fellowship
- 2012/13 Best freshman, University of Trieste
- Sep 2012–Sep 2015 Luciano Fonda BSc Fellowship